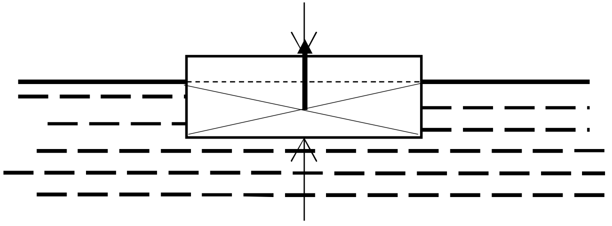
3	
(a)	Any object immersed in a fluid will experience an upthrust whose magnitude is equal to the weight of the fluid displaced by the object.
(b)(i)	Resultant force = upthrust
	= 1165.428 N = 1200 N (2sf)
(b)(ii)	 Correct point of origin and having magnitude equal to difference in length of the two arrows
(b)(iii)	By Principle of floatation, $\rho_{block} V_{block} g = \rho_{water} V_{disp} g$
	$\rho_{block} = 730 \text{ kg m}^{-3}$ (2sf)
(b)(iv)1	$\rho_{water} V_{block} g - \rho_{block} V_{block} g = \rho_{block} V_{block} a$
	$a = 4.9 \text{ m s}^{-2}$ (2sf)
(b)(iv)2	$s = \frac{1}{2} a t^2$
	= 0.20 s (2sf)
(b)(iv)3	As block moves up, viscous drag acts on the block downwards,
	reducing the net force / acceleration upwards on the block. Time taken is longer.
4	